

Lecture Notes

1 Lecture 1

Let G be a finite group acting on a finite set X .

1. Fix $g \in G$, and let $X^g = \{x \in X | gx = x\}$ be the points fixed by g acting on X . Then the number of orbits of the action of G on X is given by

$$\frac{1}{|G|} \sum_{g \in G} |X^g|$$

2. Let G act on X transitively and $|X| > 1$. Then there exists a $g \in G$ such that $g \cdot x \neq x$ for any $x \in X$.
3. Let H be a proper subgroup of G . Then there is a G action on G/H given by $(g, xH) \rightsquigarrow gxH$ (this is a transitive action).
 G is not covered by the conjugates of H , i.e., $G \neq \cup_{x \in G} xHx^{-1}$

Definition 1.1. A representation of a group G is a pair (π, V) where V is a $\mathbb{C}$ vector space and

$$\pi : G \rightarrow \text{Aut}_{\mathbb{C}}(V)$$

is a group homomorphism i.e., we have $\pi(gh) = \pi(g)\pi(h) \forall g, h \in G$.

If we fix a basis $\mathcal{B}$ of V , we get a linear map

$$\begin{aligned} \phi : \text{End}(V) &\xrightarrow{\cong} M_N(\mathbb{C}) \\ T &\mapsto [T]_{\mathcal{B}} \end{aligned}$$

Definition 1.2. If (V, π) is a representation of group G , then degree of the representation is defined as follows:

$$\deg(V, \pi) = \dim(V)$$

Example 1.3. $G = (\mathbb{Z}/n\mathbb{Z}, +)$, then

$$\chi_k : (\mathbb{Z}/n\mathbb{Z}, +) \rightarrow (S^1, \cdot) \subset (\mathbb{C}^*, \cdot)$$

defined by

$$\chi_k(\bar{1}) = e^{\frac{2\pi i}{n} k}$$

for each K such that $0 \leq k \leq n-1$. Each of the χ_k gives a one-dimensional representation of $\mathbb{Z}/n\mathbb{Z}$.

Example 1.4. Let $G = S_3$.

We get a trivial representation of S_3 given by $g \mapsto 1 \in \mathbb{C}^*$ for all $g \in S_3$.

Another one dimensional representation, *Sign representation*

$$\text{sgn} : S_3 \rightarrow \mathbb{C}^*$$

is defined as

$$g \mapsto \text{sgn}(g)$$

We define another 2-dimensional representation of S_3 . Let $S_3 = \{1, r, r^2, s, r^2s\}$.

We map

$$r \mapsto \begin{bmatrix} \cos\left(\frac{2\pi}{n}\right) & -\sin\left(\frac{2\pi}{n}\right) \\ \sin\left(\frac{2\pi}{n}\right) & \cos\left(\frac{2\pi}{n}\right) \end{bmatrix} \text{ and } s \mapsto \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

where $n = 3$.

2 Lecture 2

Suppose (π, V) is a representation of a group G . Define

$$V^* = \{l : V \rightarrow \mathbb{C} \mid l \text{ is linear}\} \subset \mathcal{F}(V)$$

Let G act on V by π , so we have $g \cdot v = \pi(g)v$. G acts on $\mathcal{F}(V)$ by the action

$$(g \cdot f)(v) = f(g^{-1} \cdot v)$$

If $l \in V^*$ we get a natural action of G on V^* defined as

$$(\pi^*(g)l)(v) = l(\pi(g^{-1}v))$$

(π^*, V^*) is called the *dual/contragredient representation* of (π, V) . Note that V^* is a G -invariant subspace of $\mathcal{F}(V)$.

Definition 2.1. Let (π, V) be a representation of a group G . We say that a subspace W of V is G -invariant if for each $g \in G$, $\pi(g)(W) \subset W$.

This gives a representation of G on W .

$$\begin{aligned} \pi|_W : G &\rightarrow \text{Aut}(W) \\ g &\mapsto \pi(g) \end{aligned}$$

$(\pi|_W, W)$ is said to be a subrepresentation of (π, V) .

Suppose (π_1, V_1) and (π_2, V_2) are representations of G . Define

$$\pi_1 \oplus \pi_2 : G \rightarrow \text{Aut}(V_1 \oplus V_2)$$

defined by

$$(\pi_1 \oplus \pi_2)(g)(v_1, v_2) = (\pi_1(g)(v_1), \pi_2(g)(v_2))$$

Let (π, V) and (τ, W) be representations of a group. We can define a representation on the vector space $\text{hom}(V, W)$. Define for $\phi \in \text{hom}(V, W)$, $g \in G$ and $v \in V$,

$$(g \cdot \phi)(v) = \tau(g)(\phi(\pi(g^{-1})(v)))$$

$$\begin{array}{ccc} V & \xrightarrow{\phi} & W \\ \downarrow g^{-1} & & \downarrow g \\ V & \xrightarrow{g\phi} & W \end{array}$$

A representation (π, V) of G is said to be irreducible if $V \neq 0$ and V has no G -invariant subspaces other than V and 0 .

Proposition 2.2. Suppose (π, v) is a representation of a finite G . Then there exists an inner product $\langle -, - \rangle$ on V such that $\langle \pi(g)v, \pi(g)w \rangle = \langle v, w \rangle \forall g \in G, v, w \in V$. (i.e.) $\langle -, - \rangle$ is a G -invariant inner product on V .

$$\langle v, w \rangle := \frac{1}{|G|} \sum_{g \in G} \langle \pi(g)v, \pi(g)w \rangle$$

Proposition 2.3. Suppose (π, v) is a representation of a finite group G . Assume that W is a G -invariant subspace of V . Then, there exists a subspace U of V such that U is G -invariant and $V = W \oplus U$.

Theorem 2.4. Let (π, V) be a finite dimensional representation of a finite group G . Then there exist irreducible subrepresentations, $W_1, \dots, W_r$ of V such that

$$V = W_1 \oplus W_2 \cdots \oplus W_r$$

3 Lecture 3

Let G be a finite group. Let (π, V) be a finite dimensional representation of the group G . Define $\chi_\pi : G \rightarrow \mathbb{C}$ by

$$\chi_\pi(g) = \text{tr}(\pi(g)).$$

Remark 3.1. Note that $g^{|G|} = e$ implies that $\pi(g)^{|G|} = I_V$. Therefore, $\pi(g)$ satisfies the polynomial $X^{|G|} - 1$, which has n distinct roots in $\mathbb{C}$. Hence there exists a basis $\mathcal{B}$ such that

$$\begin{bmatrix} \lambda_1(g) & & \\ & \ddots & \\ & & \lambda_n(g) \end{bmatrix}$$

Hence, $\text{tr}[\pi(g)]_{\mathcal{B}} = \lambda_1(g) + \cdots + \lambda_n(g)$. Since $\lambda_1, \dots, \lambda_n$ are roots of $X^{|G|} - 1$, $\lambda_1, \dots, \lambda_n \in S^1$.

Proposition 3.2. Problem (10) in Tutorial sheet.

Corollary 3.3. 1. $\chi_\pi(g^{-1}) = \chi_{\pi^*(g)} = \bar{\chi}_\pi(g)$

2. $|\chi_\pi(g)| \leq \dim(V)$.
3. $\chi_\pi(g) = \dim(V) \Leftrightarrow$ action of G on V is by scalar multiplication (that is each $\pi(g) = \alpha_g I$).
4. $\chi_\pi = \dim(V) \Leftrightarrow$ action of G on V is trivial each $\pi(g) = I$.

Let (π, V) and (τ, W) be representations on a group G . Suppose $T : V \rightarrow W$ is a linear map with

$$T(\pi(g)v) = \tau(g)(T(v)), \quad \forall g \in G, \quad v \in V.$$

The linear map T is G -linear (or G -equivariant map) if and only if, for each $g \in G$, the following diagram commutes

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \pi(g) \downarrow & & \downarrow \tau(g) \\ V & \xrightarrow{T} & W \end{array}$$

Let $\text{Hom}_G(\pi, \tau) = \{T : V \rightarrow W \mid T \text{ is } G\text{-equivariant maps}\} \subseteq \text{Hom}_{\mathbb{C}}(V, W)$ is a complex vector space.

A G -linear map T is said to be an isomorphism if T is a bijection.

Definition 3.4. Two representations (π, V) and (τ, W) on a group G are said to be *equivalent* if there exists $T \in \text{Hom}_G(\pi, \tau)$ such that T is bijective map.

Remark 3.5. 1. If T is an isomorphism, then we set from $(\star)$,

$$\tau(g) = T \circ \pi(g) \circ T^{-1}$$

for each $g \in G$.

2. $(\lambda, F(G))$ and $(\rho, F(G))$ are equivalent representations.

Lemma 3.6. Let (π, V) and (τ, W) be representations on a group G and let $T \in \text{Hom}_G(\pi, \tau)$. Then

1. $\ker(T)$ is a G -invariant subspace of V .
2. $\text{Im}(T)$ is a G -invariant subspace of W .

Lemma 3.7. Let G be a finite group. Suppose that (π, V) and (τ, W) be irreducible finite dimensional representations on G . Then

1. If π is not equivalent to τ , then $\text{Hom}_G(\pi, \tau) = \{0\}$.
2. If $V = W$, $\pi = \tau$, and $\text{Hom}_G(\pi, \tau)$, then $T = \lambda I$, for some $\lambda \in \mathbb{C}$. In fact, $\text{Hom}_G(\pi, \pi) = \{\lambda I \mid \lambda \in \mathbb{C}\}$.

Corollary 3.8. Suppose G is a finite abelian group, and (π, V) is an irreducible representation of G . Then $\dim(V) = 1$.

Theorem 3.9. Let (π, V) be a finite dimensional representation of a finite group G . Then

$$\dim(V^G) = \langle \chi_\pi, 1 \rangle = \frac{1}{|G|} \sum_{g \in G} \pi(g)$$

where V^G denotes the subspace $\{v \in V \mid \pi(g)v = v, \forall v \in V\}$.

4 Lecture 4

Proposition 4.1. Let $(\pi, V_1), (\pi_2, V_2)$ be two finite dimensional representations of G a finite group. Then

$$\langle \chi_{\pi_1}, \chi_{\pi_2} \rangle = \dim(\text{hom}_G(\pi_1, \pi_2))$$

Theorem 4.2 (Schur Orthogonality Relation). let π_1, π_2 be two irreducible representations of a finite group G . Then

$$\langle \chi_{\pi_1}, \chi_{\pi_2} \rangle = \begin{cases} 1 & \text{if } \pi_1 \text{ is equivalent to } \pi_2 \\ 0 & \text{otherwise} \end{cases}$$

Corollary 4.3. Define

$$\begin{aligned} \text{Irr}(G) &= \text{set of all irreducible representations of } G \text{ upto equivalence} \\ &= \{[\pi] | \pi \text{ irreducible representation of } G\}. \end{aligned}$$

Then $\Omega := \{\chi_\pi | \pi \in \text{Irr}(G)\}$ is a linearly independent subset of $\mathcal{F}(G)$.

Corollary 4.4. $|\text{Irr}(G)| \leq |G|$

Theorem 4.5. Let (π, V) be a finite dimensional representation of G . Let $(\pi, V) = (\tau, W_1) \oplus \dots \oplus (\tau_r, W_r)$ be a decomposition of V into a direct sum of irreducible representations. Let (ρ, W) be any irreducible representation of G , then

The number of representations (τ_j, W_j) equivalent to the representation $(\rho, W) = \langle \chi_\pi, \chi_\rho \rangle$

Corollary 4.6. Suppose (π_1, V_1) and (π_2, V_2) are two finite dimensional representations of a finite group G . Then the following are equivalent:

1. (π_1, V_1) is equivalent to (π_2, V_2)
2. $\chi_{\pi_1} = \chi_{\pi_2}$

Theorem 4.7. Let $\text{Irr}(G)$ denote the set of inequivalent irreducible representations of a finite group G .

$$1. \chi_\lambda = \chi_\rho = \sum_{\pi \in \text{Irr}(G)} \chi_\pi(e) \chi_\pi = \sum_{\pi \in \text{Irr}(G)} \dim(\pi) \chi_\pi.$$

$$2. |G| = \sum_{\pi \in \text{Irr}(G)} (\chi_\pi(e))^2 = \sum_{\pi \in \text{Irr}(G)} (\dim(\pi))^2.$$

$$3. \sum_{\pi \in \text{Irr}(G)} \chi_\pi(1) \chi_\pi(g) = 0 \text{ for each } g \in G \text{ such that } g \neq e.$$

Definition 4.8 (Center of $\mathcal{F}(G)$). $Z(\mathcal{F}(G)) = \{f \in \mathcal{F}(G) | f * \phi = \phi * f \forall \phi \in \mathcal{F}(G)\}$

Proposition 4.9. $f \in Z(\mathcal{F}(G)) \iff f$ is a class function

$Z(\mathcal{F}(G))$ is a subspace of $\mathcal{F}(G)$. Let $\{C_1, \dots, C_r\}$ denote the conjugacy classes in G . Define

$$1_{C_i} : G \rightarrow \mathbb{C} \quad 1_{C_i}(g) = \begin{cases} 1 & g \in C_i \\ 0 & \text{otherwise} \end{cases}$$

$\{1_{C_1}, \dots, 1_{C_r}\}$ form a linearly independent set and span $Z(\mathcal{F}(G))$.

So $\dim(Z(\mathcal{F}(G))) = \text{number of conjugacy classes in } G$.

Problems

1. Suppose G acts on X . Let $\mathcal{F}(X) := \{f : X \rightarrow \mathbb{C}\}$. Show that G acts on $\mathcal{F}(X)$ as follows:
 $g \in G, x \in X \quad (g \cdot f)(x) = f(g^{-1} \cdot x)$
2. Suppose X is a finite set. Consider $(\mathcal{F}(X), +, \cdot)$ where $+$ is pointwise addition and $\cdot$ is scalar multiplication
 - (a) Show that $(\mathcal{F}(X), +, \cdot)$ is a $\mathbb{C}$ -vector space. What is the $\dim(\mathcal{F}(X))$? Can you give a basis of $\mathcal{F}(X)$?
 - (b) Is the action of G on $\mathcal{F}(X)$ linear? If so, what is the representation of G on $\mathcal{F}(X)$ obtained via this action?
3. G acts on G via:
 - (i) left action: $\lambda(g)x = g \cdot x$
 - (ii) right action: $\rho(g)x = x \cdot g^{-1}$

From Problem 1, G acts on $\mathcal{F}(G)$ via λ and ρ . What are the actions and write down the representations: $\lambda : G \rightarrow \text{Aut}(\mathcal{F}(G))$ and $\rho : G \rightarrow \text{Aut}(\mathcal{F}(G))$.

4. Using Problem 3, i.e. λ and ρ , we can define action of $G \times G$ on G ; $(g, h) \cdot x = gxh^{-1}$, $g, x, h \in G$. What is the induced action of $G \times G$ on $\mathcal{F}(G)$?
5. Assume that G is finite. On $\mathcal{F}(G)$, define a new operation $\star$ as follows:

$$f_1, f_2 \in \mathcal{F}(G) \quad f_1 \star f_2 : G \rightarrow \mathbb{C} \quad \text{defined by} \\ (f_1 \star f_2)(x) = \frac{1}{|G|} \sum_{g \in G} f_1(g)f_2(g^{-1}x)$$

- (a) Show that $\star$ is associative.
 - (b) Show that $(\mathcal{F}(G), +, \cdot, \star)$ is a $\mathbb{C}$ -algebra.
 - (c) Does $(\mathcal{F}(G), +, \star)$ have an identity element?
6. Prove that sign representation $g \rightsquigarrow \text{sign } g$ and the trivial representation are the only one-dimensional representations of the symmetric group S_n .

7. (a) If X is any finite set and G acts on X . Then there is an associated permutation representation: Let V be the vector space with basis $\{e_x : x \in X\}$ and let G act on V by

$$g \cdot \sum_{x \in X} a_x e_x = \sum_x a_x e_{gx}$$

The *left(right) regular representation* denoted R_G or R corresponds to the left (right) action of G on itself.

- (b) Recall the vector space $\mathcal{F}(G)$ and the G action on this vector space. This is an alternate description of the regular representation.

Verify that these two descriptions of R agree.

8. If V is G representation, then we have a map $\rho : G \rightarrow \text{Aut}(V)$. Note that this map in general is not a G -module homomorphism (or G -linear: for $h \in G$

$$h \cdot g(v) \neq g(h \cdot v)$$

Prove that $\rho(g) : V \rightarrow V$ is G -linear for every ρ if and only if g is in the center $Z(G)$ of G .

9. (i) If G is abelian and V is an irreducible representation of G , use Schur's lemma to describe the action of g on V .
(ii) What would be the dimension of V ?
(iii) Describe all the irreducible representations of G .
10. Describe the character table for the groups: $\mathbb{Z}_n, \mathbb{Z}_2 \times \mathbb{Z}_2, A_4, S_3$.